

Experiential Educational Accessibility Modules

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ABSTRACT



Our *Accessibility Learning Labs* both inform participants about how to properly create accessible software, and also demonstrate the need to create accessible software. Five labs are currently available on the topics of: Colorblindness, Hearing, Blindness and Dexterity. Material is available on our website: <https://all.rit.edu>

Due to their self-contained web-based nature and the inclusion of all instructional materials (e.g., slides, quizzes, etc.), the labs enable easy integration into a wide variety of curricula ranging from high schools (9-12) to graduate courses. This session will provide an overview of the labs, and usage instructions and information for adopters.

CCS CONCEPTS

• **Human-centered computing** → *Accessibility*; • **Social and professional topics** → *CS1*; *Software engineering education*.

KEYWORDS

Accessibility Education, Computing Education, Computing Accessibility

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1 BENEFITS

While creating accessible software is imperative for making software inclusive for all users, problematically the topic of accessibility is frequently excluded from computing education. This leads to scenarios where students are not only unaware of how to create

accessible software, but do not see the need to create accessible software. To address this challenge, we have created a set of educational *Accessible Learning Labs* (ALL) that are systematically designed to not only inform students about fundamental topics in computing accessibility, but to demonstrate the importance of creating accessible software as well.

1.1 Lab Structure, Topics and Goals

1.1.1 Lab Goals. The educational accessibility labs have several primary goals:

- (1) *The labs do not require special software or hardware:* Easy adoption of the material is supported by the fact that all material is hosted on our project servers, and requires only a browser and internet connection for usage.
- (2) *Adopters need only basic programming skills:* Students possessing a diverse set of experience levels and skillsets should be made aware of the importance of creating accessible software and be provided foundational skills to create accessible software.
- (3) *The labs should fit into already crowded computing courses:* Each lab is designed to take approximately 30-60 minutes, and the instructor may select the lab components that they would like to utilize in an à-la-carte fashion inside or outside of the classroom. The succinctness of the labs will enable them to fit into courses that are already heavily time-constrained.
- (4) *The labs include all instructional content necessary for adoption:* To support instructors and participants to easily utilize the created material, all necessary content (e.g., lecture slides, activity, quiz, etc.) are provided.
- (5) *The labs should demonstrate the importance of creating accessible software:* Teaching participants about how to create accessible software is important, but participants should also be made aware *why* it's essential to create accessible software.

The labs enable participants to directly experience both the impact of inaccessible software and the impact that their modifications have on making the software more accessible. As an example, for ensuring that software is accessible to users with hearing impairment, a common best practice will be to not rely solely upon audio cues since this can adversely impact the experience of Deaf/Hard of Hearing users — participants could repair the application through the use of text or visual aides in addition to any audio notifications [2]. Figure 1 provides an example of this process. Figure 1a

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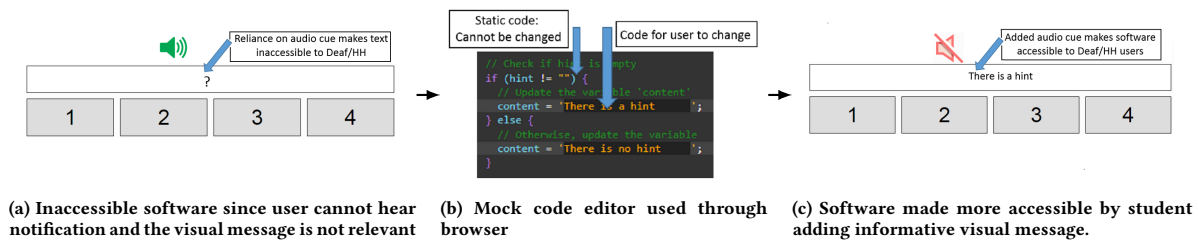


Figure 1: Example of student repairing accessibility problem using simulated code editor

demonstrates the inaccessible version of the software, while Figure 1b shows the simulated code editor to change the source code, and Figure 1c demonstrates the repaired, accessible version of the application.

1.1.2 *Lab Structure.* Each lab contains the components shown in Table 1

2 INTENDED AUDIENCE(S)

Anyone in the software engineering community who would like to ensure that they are properly creating inclusive and accessible software will be able to learn and benefit from this course, varying from first-year students to experienced practitioners. Instructors of computing and non-computing courses who are interested in including foundational accessibility concepts. While the labs are specifically targeted towards undergraduate introductory computing courses (CS1 & CS2), they may be used in a variety of other courses ranging from 9-12 grade high school courses, to upper level undergraduate courses, and even in graduate-level curriculum.

The session will provide participants with a mechanism to I) Learn about our provided educational accessibility material, II) Provide feedback and guidance on any necessary modifications to the labs, and III) Share common challenges and best practices for including accessibility in computing education. Session participants may choose to participate by discussing and reflecting on included topics, or may use their own laptop to experience the material in a hands-on manner. No prior input from event participants will be required.

3 PREREQUISITES

While general knowledge on the topic of computing accessibility will be helpful, there is no significant *required* background for this Course.

4 CONTENT

The session will include the following items (75 minutes):

4.1 Session Agenda

- (1) **Activity 1: Overview of Labs (10 minutes):** An overview of the labs, their objectives and their components.
- (2) **Activity 2: Lab # 1: Hearing (20 minutes):**
 - **Learning Objective:** Introduce the concept of making software accessible to users who are Deaf or Hard of Hearing (DHH).

- **Difficulty Level (for students):** Introductory

- **Activity:** This lab involves participants playing a game where they are tasked with locating a random, hidden item. Points are awarded for finding the item quickly. An audio cue randomly provides the location of the hidden item, thus enabling the user to identify it sooner with more accuracy and achieve a higher score. The accessibility emulation component involves simply not playing the audio cue. To make the software more accessible, the participant adds a visual cue for the hint thus making the software accessible to DHH users. This lab was created by a DHH student.

(3) Activity 3: Lab # 2: Colorblindness (20 minutes):

- **Learning Objective:** The primary learning objective of this lab is to inform students about the *Distinguishable Content* accessibility guideline.
- **Difficulty Level (for students):** Introductory
- **Activity:** This lab introduces the concept of making software accessible to users with colorblindness. In this game participants are asked to click on specifically colored circles when they appear on the screen. After the first round of the game, a color-blindness emulation feature is activated which makes the colors appear as they would to someone who has colorblindness. This makes the task of clicking on specifically defined colors very difficult, meaning that most of the 'unsafe' colors that are used are indistinguishable from one another. Participants are tasked with using proper colors to make the software accessible to users with colorblindness. This is accomplished by having the participants modifying the colors used in the application, and then using the software with the colorblindness emulation feature still active.

- (4) **Activity 4: Lab Discussion (10 minutes):** We will discuss the labs that were presented along with the future of the project. This will include: A) What new labs should we be creating and what topics are most imperative for instructors to be addressing B) The alterations that need to be made to existing labs C) How to make our material more effective and easily adoptable.
- (5) **Activity 5: Accessibility Education Discussion (10 minutes):** This will provide a mechanism for participants to discuss common challenges and advice for including accessibility in computing education.

Table 1: Lab Components

Component	Description
Accessibility information	Background information on the accessibility challenge being addressed. This will include compliance information and information about how users are affected.
Lab instructions	Information how to complete the labs.
Lecture slides	Example lecture slides that an instructor may use to present the material. The slides will be in ppt format so instructors can still alter them as they'd like.
Video presentation of lecture slides	ADA Compliant YouTube video of project member presenting the lecture. This will support students conducting the labs on their own, or the flipped classroom environment.
Empathy-creating material	Written and ADA compliant YouTube video from actual people with examined people with disabilities.
Video of lab being conducted	ADA compliant YouTube video of the lab being conducted.
Lab activity	The lab activity to be conducted.
Emulation component	Feature to emulate access challenges for people with disabilities.
Ethics	Ethical and legal importance of developing accessible software.
Quiz	An example quiz will be provided to instructors if they would like to evaluate their students at the conclusion of the activity.

- (6) **Activity 6: Conclusion/Q&A (5 minutes):** Recap of the labs and a question and answer period.

4.2 Session Relevance

Our world needs to create more accessibility inclusive software. Therefore, we need an accessibility literate workforce that not only understands how to create accessible software, but also recognizes the importance of creating accessible software. Unfortunately, although accessibility is a crucial computing topic, it is often excluded from education [1, 3]. To support the inclusion of this imperative computing topic, we need high-quality, easily adoptable material for our computing courses. Our previous publications have demonstrated the effectiveness of both experiential educational interventions, and the pedagogical effectiveness of the ALL material and their format [4, 5].

5 PRACTICAL WORK

We will be asking participants to go through the different components of the lab which includes the background material, an interactive exercise which simulates the accessibility issue which the participant will learn how to fix, reinforcement materials mainly consisting of video testimonials of people's real life experiences regarding the issue, and a quiz where the participant will be able to test the knowledge that they have gathered throughout the lab.

6 INSTRUCTOR BACKGROUND

Samuel Malachowsky is the Co-PI of the project that is developing the described accessibility labs. Malachowsky is a Sr. Lecturer at RIT and has authored 9 pedagogically focused publications. He holds a Project Management (PMP) certification and has authored a textbook on team leadership.

Daniel Krutz is the PI of the NSF-funded project [4, 5] that is devoted to creating the presented labs. Krutz has taught approximately ten different graduate and undergraduate software engineering courses and is the author of over twelve pedagogical research papers.

7 RESOURCES

Complete Course materials can be found on our website: <https://all.rit.edu>.

8 ACCESSIBILITY

Complete instructional materials are already ADA compliant. Interpreters will be provided as requested per conference guidelines.

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